Tetris

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Motivation

MODERN GAMES RUN ON SOFTWARE PROCESSORS

HARDWARE-ONLY SYSTEMS ARE FASTER

NEED FOR REAL-TIME EMBEDDED SYSTEMS

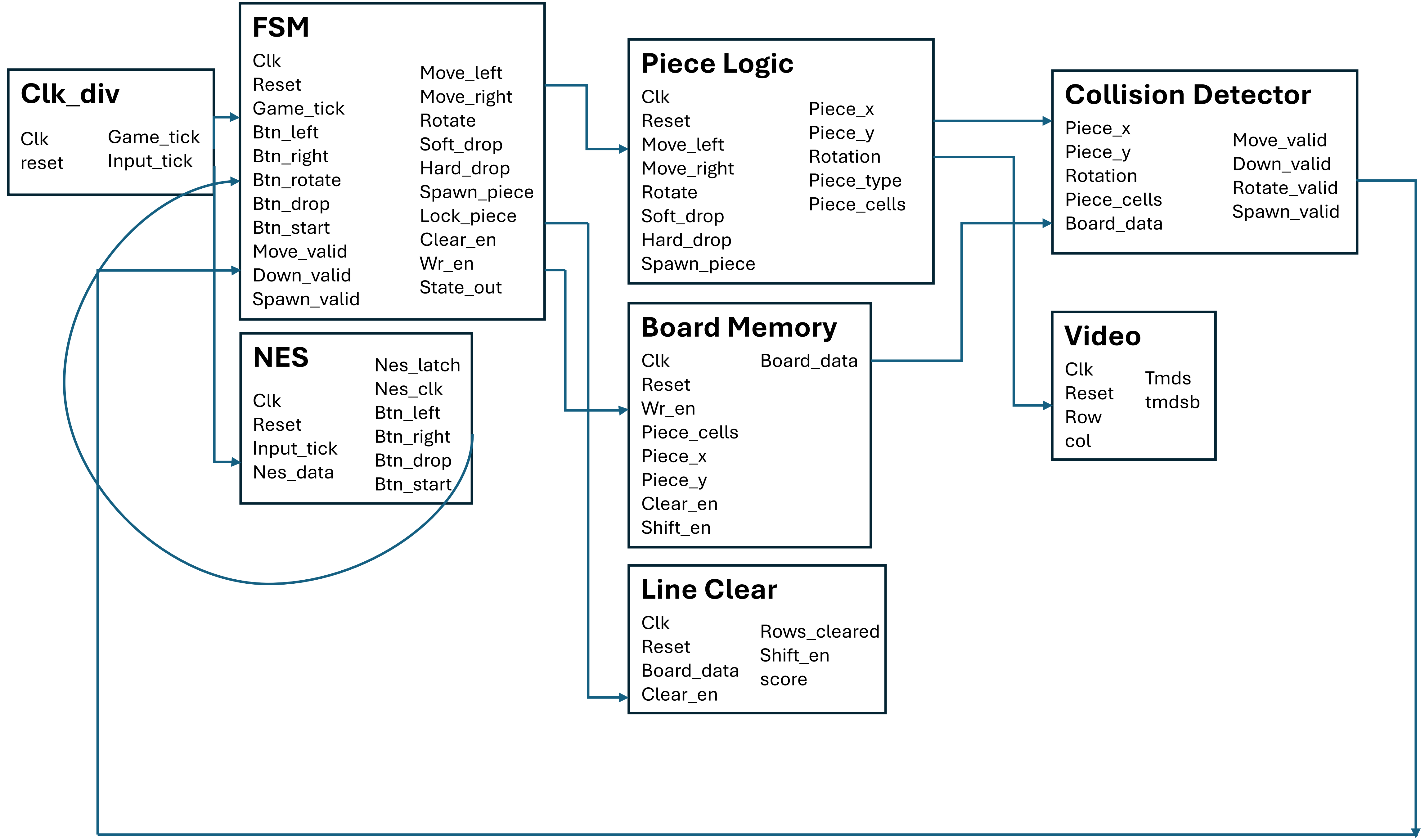
Technical Objectives

Implement Tetris in pure VHDL

Real-time video output

Hardware FSM game control

External controller


Clk_div
Clk
reset
Game_tick
Input_tick
FSM
Clk
Reset
Game_tick
Btn_left
Btn_right
Btn_rotate
Btn_drop
Btn_start
Move_valid
Down_valid
Spawn_valid
Move_left
Move_right
Rotate
Soft_drop
Hard_drop
Spawn_piece
Lock_piece
Clear_en
Wr_en
State_out
NES
Clk
Reset
Input_tick
Nes_data
Nes_latch
Nes_clk
Btn_left
Btn_right
Btn_drop
Btn_start
Piece Logic
Clk
Reset
Move_left
Move_right
Rotate
Soft_drop
Hard_drop
Spawn_piece
Piece_x
Piece_y
Rotation
Piece_type
Piece_cells
Board Memory
Clk
Reset
Wr_en
Piece_cells
Piece_x
Piece_y
Clear_en
Shift_en
Board_data
Line Clear
Clk
Reset
Board_data
Clear_en
Rows_cleared
Shift_en
score
Collision Detector
Piece_x
Piece_y
Rotation
Piece_cells
Board_data
Move_valid
Down_valid
Rotate_valid
Spawn_valid
Video
Clk
Reset
Row
col
Tmds
tmdsb

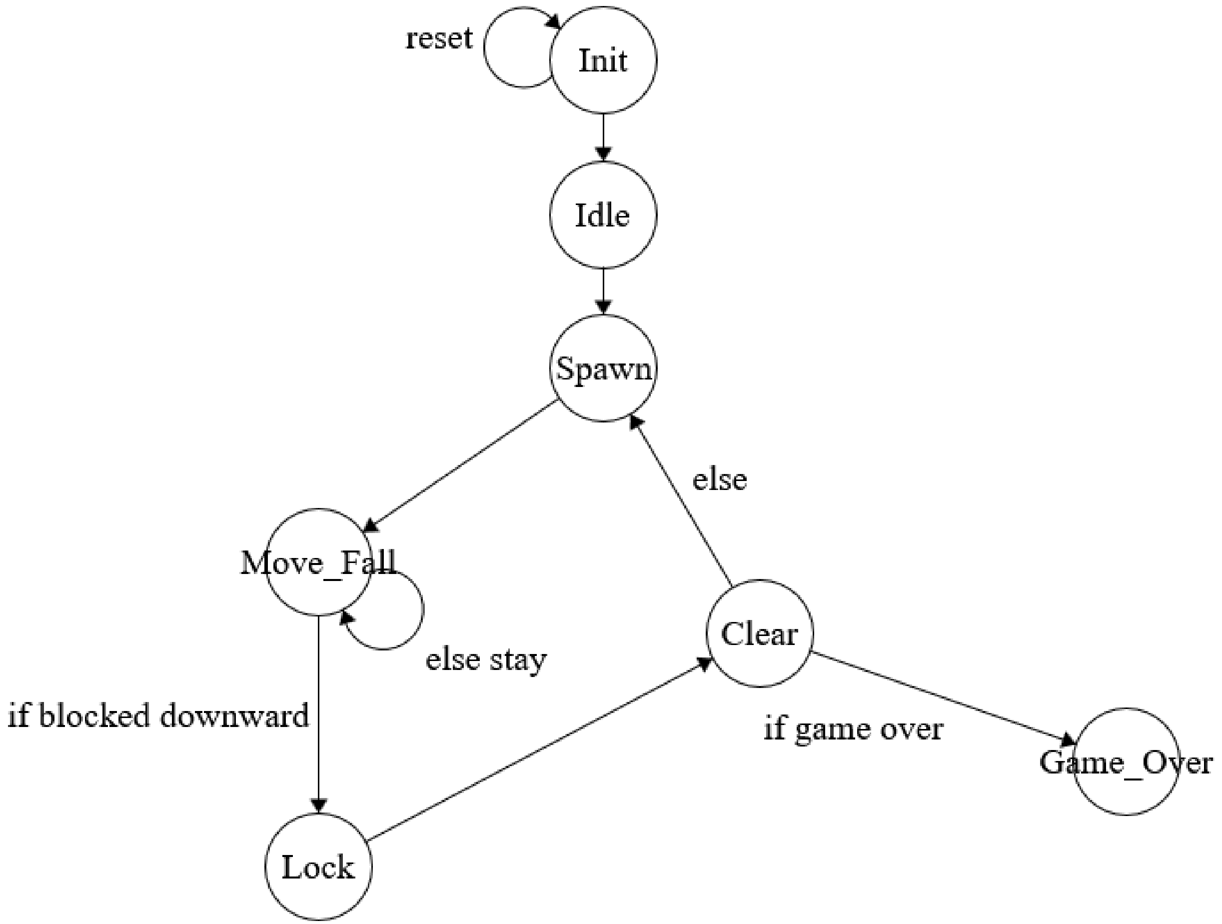
Data Flow

Inputs to FSM decisions

FSM to movement commands

Board updates, rendered by frame

FSM


reset
Init
Idle
Spawn
else
Move_Fall
else stay
Clear
if blocked downward
if game over
Game_Over
Lock

Piece Logic

- Tracks piece position
- Handles left/right movement, rotation, drops

Piece Logic

Inputs	Outputs
Clk	Piece_x
Reset	Piece_y
Move_left	Rotation
Move_right	Piece_type
Rotate	Piece_cells
Soft_drop	
Hard_drop	
Spawn_piece	

Collision Detector

- Checks move validity
- Outputs valid signal
- Computes hard drop

Collision Detector

Piece_x
Piece_y
Rotation
Piece_cells
Board_data

Move_valid
Down_valid
Rotate_valid
Spawn_valid

Board Memory

- Stores 10x20 grid as array
- Writes only on lock
- Handles moving pieces down after line clear

Board Memory

Inputs	Outputs
Clk	Board_data
Reset	
Wr_en	
Piece_cells	
Piece_x	
Piece_y	
Clear_en	
Shift_en	

Line Clear

- Detects full rows
- Outputs rows_cleared
- Keeps track of score
- FSM Handles shift after clear

Line Clear

Inputs	Outputs
Clk	Rows_cleared
Reset	Shift_en
Board_data	score
Clear_en	

Issues!

- False game over, add wait states!
- Random Colors, rework board memory
- Hard drop glitches, move logic to collision detector
- Controller issues

Questions?